

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Provisioned and operated multi-cloud GPU clusters (AWS, GCP, CoreWeave) with NVIDIA/CUDA hardware and InfiniBand/RoCE networking, supporting long-running PyTorch training jobs and high-availability inference on shared fleet capacity.
- Wrote infrastructure/systems-level code in C++, Go, and Rust for scheduling, resource allocation, and fault-tolerant placement logic under GPU-scarcity constraints.
- Applied model compression and quantization to run PyTorch models in lower precision for efficient inference, reducing serving cost without regressing accuracy.
- Built observability tooling tracking GPU utilization, fault detection, and job health across the training and inference fleet, replacing polling with event-driven monitoring.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Extended and optimized vLLM-class inference framework internals with CUDA/Triton kernel-level work, distributed KV cache handling, and speculative decoding to cut serving latency.
- Benchmarked 5+ multi-agent RL architectures on a multi-GPU PyTorch cluster under distributed inference for robustness-critical pipelines.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards across the team.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus key-value store in C++ with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing for fault-tolerant distributed coordination.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, C++, Go, Rust, Java

ML & Inference: PyTorch, Model Training/Fine-Tuning, Model Compression, Quantization, CUDA/Triton Kernels, vLLM

GPU & Distributed Systems: NVIDIA/CUDA, InfiniBand/RoCE, Multi-Cloud GPU Orchestration, Fault-Tolerant Scheduling

Cloud & Infra: AWS, GCP, CoreWeave, Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer